

CI/CD: Automated AWS EC2 Deployment via GitHub Actions

STEP 1

ubuntu@ip-172-31-33-10:~\$

```
>cd ~/nest-app-dep
>nano deploy.sh
```

Paste this below code

```
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
```

```
#!/bin/bash
```

```
set -e
```

```
export NVM_DIR="$HOME/.nvm"
```

```
[ -s "$NVM_DIR/nvm.sh" ] && \. "$NVM_DIR/nvm.sh"
```

```
cd ~/nest-app-dep
```

```
echo "Pulling latest code..."
```

```
git pull origin main
```

```
echo "Installing dependencies..."
```

```
npm ci
```

```
echo "Building..."
```

```
NODE_OPTIONS="--max-old-space-size=2048" npm run build
```

```
echo "Restarting app..."
```

```
pm2 restart nest-blog-prod
```

```
echo "Deploy complete."
```

```
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
```

```
Ctrl o => enter => ctrl x (file saved)
```

Verify to run this script

```
>bash ~/nest-app-dep/deploy.sh
```

STEP 2 Generate a dedicated deploy SSH key

ubuntu@ip-172-31-33-10:~\$

```
>ssh-keygen -t ed25519 -f ~/.ssh/deploy_key -N ""  
cat ~/.ssh/deploy_key.pub >> ~/.ssh/authorized_keys  
cat ~/.ssh/deploy_key
```

Example: copy it or save on notepad

-----BEGIN OPENSSH PRIVATE KEY-----

```
b3BlbnNzaC1rZXktdjEAAAABG5vbmUAAAAEbm9uZQAAAAAAAAABAAAAMwAAAAAtzc2gtZWQyNTUxOQAAACcvENoTuLX8Az8QzRDbcCEnO7jtG5NwOBhHy4KpikcwEgAAAKAdbKI4HWYi  
i  
OAAAAAtzc2gtZWQyNTUxOQAAACcvENoTuLX8Az8QzRDbcCEnO7jtG5NwOBhHy4KpikcwEg  
g  
AAAEbfoINKvRHICiVtx3yuMeid1bKkTCvaN3P9sst2amXnGK8Q2hO4tfwDPxDNENtwlSc7  
uO0bk3A4GEfLgqmKRzASAAAFnVidW50dUBpcC0xNzltMzEtMzMtMTABAgMEBQYH  
-----END OPENSSH PRIVATE KEY-----
```

STEP 3 Add secrets in GitHub

In your repo: Settings → Secrets and variables → Actions → New repository secret, add:

SSHPRIVATEKEY — ssh-key

SSHHOST — your Elastic IP (172.31.33.10 or its public IP)

SSHUSER — ubuntu

STEP 4

In your repo, create .github/workflows/deploy.yml:

```
name: Deploy  
  
on:  
  push:  
    branches: [main]
```

```
jobs:
  deploy:
    runs-on: ubuntu-latest
    steps:
      - name: Deploy to server
        uses: appleboy/ssh-action@v1.0.3
        with:
          host: ${ secrets.SSHHOST }
          username: ${ secrets.SSHUSER }
          key: ${ secrets.SSHPRIVATEKEY }
          script: bash ~/nest-app-dep/deploy.sh
```